

Theorem 1 – Formal Setup

Direction-Invariant Label Flip under Linear Batch Correction

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1 Objects and notation

Let

$X \in \mathbb{R}^{n \times p}$	row-wise expression matrix,
$Y \in \{0, 1\}^n$	binary clinical label,
$B \in \{1, \dots, K\}^n$	cohort (batch) assignment, $K \geq 2$,
$\pi_k = \{i : B_i = k\}$	index set of samples in batch k .

All rows X_i are assumed column-standardised across the full dataset so that sample-level covariance is order-one.

1.1 Key directions

We define the three unit vectors that drive the analysis.

$$w_Y = \frac{\mu_1 - \mu_0}{\|\mu_1 - \mu_0\|}, \quad \mu_c = \mathbb{E}[X_i \mid Y_i = c] \quad (c \in \{0, 1\}),$$

(label direction)

$$w_B = \arg \max_{\|w\|=1, w \perp \ker(P_Y^\perp)} w^\top \text{Cov}_B(X) w, \quad \text{Cov}_B(X) = \mathbb{E}_k[(\mu_{B=k} - \bar{\mu})(\mu_{B=k} - \bar{\mu})^\top],$$

(dominant batch direction)

$$w_{\text{trained}} = \arg \min_{w, b} \sum_{i=1}^n \ell(Y_i, w^\top T_B(X)_i + b) + \lambda \|w\|^2, \quad \text{(trained classifier)}$$

where ℓ is a convex margin loss (logistic, hinge, or squared-hinge) and $\lambda \geq 0$ is a ridge penalty.

We write $\rho = \langle w_Y, w_B \rangle \in [-1, 1]$ and note that under the sign convention above $\rho \geq 0$ after at most a relabelling of the batch coordinate.

1.2 Projection operators

Definition 1 (Batch subspace projector). $P_B = U_B U_B^\top$, where $U_B \in \mathbb{R}^{p \times r}$ has orthonormal columns spanning the column space of the centred batch-indicator matrix $\tilde{B} \in \mathbb{R}^{n \times K}$ acting on X ; i.e. $\text{range}(P_B) = \text{span}\{\mu_{B=k} - \bar{\mu} : k = 1, \dots, K\}$. Rank $r = K - 1$ generically.

Definition 2 (Label direction projector). $P_Y = w_Y w_Y^\top$.

1.3 Linear batch correction with label covariate

Following the standard ComBat-with-covariate formulation, we model

$$X_i = \alpha + \beta^\top Y_i + \gamma_{B_i} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \Sigma). \quad (1)$$

The linear correction operator is

$$T_B(X_i) = X_i - \hat{\gamma}_{B_i} = X_i - (\bar{X}_{\pi_{B_i}} - \bar{X} - \hat{\beta}^\top (\bar{Y}_{\pi_{B_i}} - \bar{Y})), \quad (2)$$

where $\hat{\gamma}_k$ is the OLS estimate of γ_k holding the label covariate Y fixed. Matrix form: there exists $M_B \in \mathbb{R}^{n \times n}$ (idempotent on the batch-indicator column space, orthogonal to Y) such that

$$T_B(X) = (I - M_B)X + (\text{label-covariate adjustment}). \quad (3)$$

1.4 The key quantity

The *alignment parameter*

$$\rho^2 = \langle w_Y, w_B \rangle^2 \in [0, 1] \quad (4)$$

measures how closely the dominant batch direction parallels the true label direction. **Theorem 1** (stated below) shows that $\rho^2 > \frac{1}{2}$ is the sharp threshold at which the trained classifier under T_B *inverts* relative to the pre-correction discriminant.

2 Assumptions

Assumption 1 (Additive generative model). *Equation (1) holds; residuals are isotropic with sub-Gaussian tails.*

Assumption 2 (Well-posedness). $\mu_1 \neq \mu_0$ and at least two batches differ in mean; the ridge parameter satisfies $\lambda = o(n)$.

Assumption 3 (Cohort imbalance or near-confounding). *There exist $k \neq k'$ with $|\Pr(Y = 1 \mid B = k) - \Pr(Y = 1 \mid B = k')| \geq \delta > 0$. This is what makes ρ strictly positive in the finite-sample regime.*

Assumption 4 (Sample scaling). n/p is bounded away from zero as $n \rightarrow \infty$; $\min_k |\pi_k| = \Omega(n/K)$.

3 Main theorem (statement)

Theorem 1 (Direction-invariant label flip). *Assume 1-4. Let $f_{\text{trained}}(x) = w_{\text{trained}}^\top x + b_{\text{trained}}$ be the linear classifier fit on $(T_B(X), Y)$. Let*

$$\text{AUC}_{\text{post}} = \Pr(f_{\text{trained}}(X^*) > f_{\text{trained}}(X^\dagger) \mid Y^* = 1, Y^\dagger = 0)$$

on a held-out cohort, and set

$$\text{DIAL} = \max(\text{AUC}_{\text{post}}, 1 - \text{AUC}_{\text{post}}) - \text{AUC}_{\text{post}}.$$

Then there exists a universal threshold $\rho^ \in (0, 1)$ with $\rho^{*2} = \frac{1}{2}$ such that*

- (i) *if $\rho^2 < \rho^{*2}$, $\lim_{n \rightarrow \infty} \Pr(\text{AUC}_{\text{post}} > 1/2) = 1$, and $\text{DIAL} \xrightarrow{p} 0$;*
- (ii) *if $\rho^2 > \rho^{*2}$ and Assumption 3 holds, $\lim_{n \rightarrow \infty} \Pr(\text{AUC}_{\text{post}} < 1/2) = 1$, and $\text{DIAL} \xrightarrow{p} 1 - 2 \text{AUC}_{\text{post}}^\infty > 0$.*

Equivalently, $\text{sign}\langle w_{\text{trained}}, w_Y \rangle$ flips sharply at $\rho^2 = \frac{1}{2}$.

Corollary 2 (DIAL characterisation). *Under Theorem 1, $\text{DIAL} > 0$ if and only if $\rho^2 > \frac{1}{2}$. In particular, DIAL is a direct estimator of the mass on the $\rho^2 > \frac{1}{2}$ event conditional on cohort structure.*